Name: Noorain Patankar
Roll no: 5855
SYBCA "A"

PRACTICAL NO 8

1. Design database for Employee management system. Develop JSP(s) to perform CRUD operations.

database and table:

```
mysql> create database emedb;
Query OK, 1 row affected (0.01 sec)

mysql> use emedb;
Database changed
```

```
mysql> CREATE TABLE employee (
    ->     emp_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     name VARCHAR(100) NOT NULL,
    ->     department VARCHAR(100),
    ->     salary DOUBLE
    -> );
Query OK, 0 rows affected (0.05 sec)
```

CODE:

db.jsp

```
<%@ page import="java.sql.*" %>
<%
    String url = "jdbc:mysql://localhost:3306/emedb";
    String user = "root";
    String pass = "mamta123";
    Class.forName("com.mysql.cj.jdbc.Driver");
    Connection con = DriverManager.getConnection(url, user, pass);
%>
```

addEmployee.jsp

```
<%@ include file="db.jsp" %>
<%
    String name = request.getParameter("name");
    String dept = request.getParameter("dept");
    String salary = request.getParameter("salary");
    PreparedStatement ps = con.prepareStatement(
        "insert into employee(name, department, salary) values(?, ?, ?)"
    );
    ps.setString(1, name);
    ps.setString(2, dept);
```

```
    ps.setDouble(3, Double.parseDouble(salary));
    ps.executeUpdate();
    response.sendRedirect("viewEmployee.jsp");
%>
```

deleteEmployee.jsp

```
<%@ include file="db.jsp" %>
<%
    int id = Integer.parseInt(request.getParameter("id"));
    PreparedStatement ps = con.prepareStatement(
        "delete from employee where emp_id = ?"
    );
    ps.setInt(1, id);
    ps.executeUpdate();
    response.sendRedirect("viewEmployee.jsp");
%>
```

index.jsp

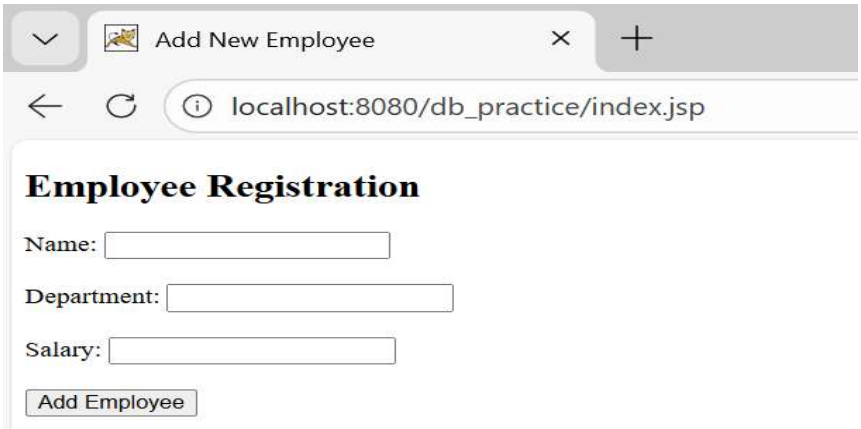
```
<!DOCTYPE html>
<html>
<head>
    <title>Add New Employee</title>
</head>
<body>
    <h2>Employee Registration</h2>
    <form action="addEmployee.jsp" method="get">
        Name: <input type="text" name="name" required><br><br>
        Department: <input type="text" name="dept" required><br><br>
        Salary: <input type="number" name="salary" step="0.01" required><br><br>
        <input type="submit" value="Add Employee">
    </form>
</body>
</html>
```

viewEmployee.jsp

```
<%@ include file="db.jsp" %>
<table border="1">
```

```
    <tr>
        <th>ID</th>
        <th>Name</th>
        <th>Dept</th>
        <th>Salary</th>
        <th>Action</th>
    </tr>
<%
    Statement st = con.createStatement();
    ResultSet rs = st.executeQuery("select * from employee");
    while(rs.next()) {
%>
    <tr>
        <td><%= rs.getInt("emp_id") %></td>
        <td><%= rs.getString("name") %></td>
        <td><%= rs.getString("department") %></td>
        <td><%= rs.getString("salary") %></td>
        <td>
            <a href="deleteEmployee.jsp?id=<%= rs.getInt("emp_id") %>">Delete</a>
        </td>
    </tr>
<%
    }
%>
</table>
```

OUTPUT:



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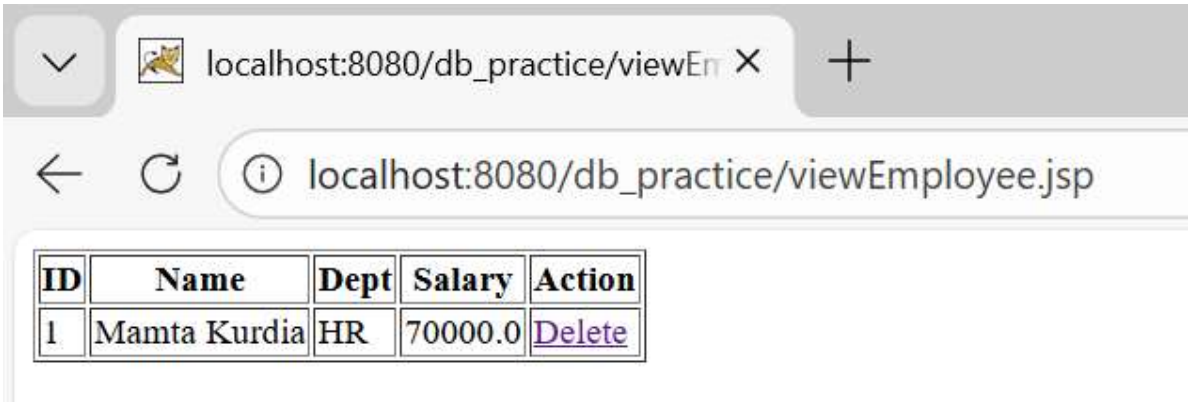
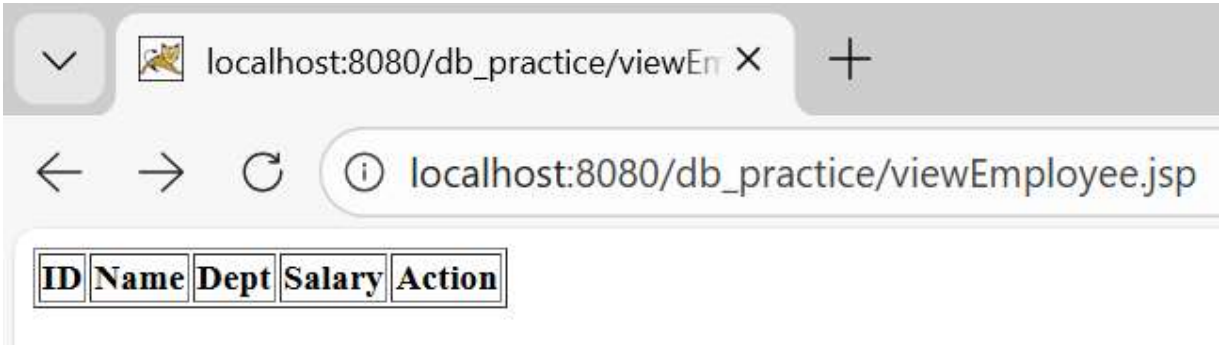
After adding Employee:



ID	Name	Dept	Salary	Action
1	Mamta Kurdia	HR	70000.0	Delete

```
mysql> select * from employee;
+--------+--------------+------------+--------+
| emp_id | name         | department | salary |
+--------+--------------+------------+--------+
|      2 | Mamta Kurdia | HR         |  70000 |
+--------+--------------+------------+--------+
1 row in set (0.02 sec)
```

After deleting Employee:



ID	Name	Dept	Salary	Action

```
mysql> select * from employee;
Empty set (0.00 sec)
```

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2. Design database for student administration. Develop servlet(s) to perform CRUD operations.

Database and table:

```
mysql> CREATE DATABASE studentdb;
Query OK, 1 row affected (0.02 sec)

mysql> USE studentdb;
Database changed
mysql> CREATE TABLE student (
    -> sid INT PRIMARY KEY AUTO_INCREMENT,
    -> name VARCHAR(100),
    -> course VARCHAR(100),
    -> marks INT
    -> );
Query OK, 0 rows affected (0.14 sec)
```

CODE:

index.html

```
<form action="StudentServlet" method="post">
   Name: <input type="text" name="name"><br>
   Course: <input type="text" name="course"><br>
   Marks: <input type="text" name="marks"><br>
   <input type="submit" value="Save">
</form>
```

StudentServlet.java

```
import java.io.*;
import java.sql.*;
import jakarta.servlet.*;
import jakarta.servlet.http.*;
public class StudentServlet extends HttpServlet {
Connection con;
public void init() {
try {
Class.forName("com.mysql.cj.jdbc.Driver");
con = DriverManager.getConnection(
"jdbc:mysql://localhost:3306/studentdb","root","mamta123");
} catch(Exception e) {}
}
```

```
protected void doPost(HttpServletRequest req,
HttpServletResponse res) throws ServletException,
IOException {
String name = req.getParameter("name");
String course = req.getParameter("course");
int marks = Integer.parseInt(req.getParameter("marks"));
try {
PreparedStatement ps = con.prepareStatement( "INSERT INTO student(name,course,marks) VALUES(?,?,?)");
ps.setString(1,name);
ps.setString(2,course);
ps.setInt(3,marks);
ps.executeUpdate();
res.sendRedirect("success.html");
} catch(Exception e) {}
}
}
```

success.html

```
<!DOCTYPE html>
<html>
<head>
   <title>Success</title>
</head>
<body>
   <h2>Student data saved successfully!</h2>
   <a href="index.html">Add Another Student</a>
</body>
</html>
```

web.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/javaee
        http://xmlns.jcp.org/xml/ns/javaee/web-app_4_0.xsd"
        version="4.0">


<servlet>
```

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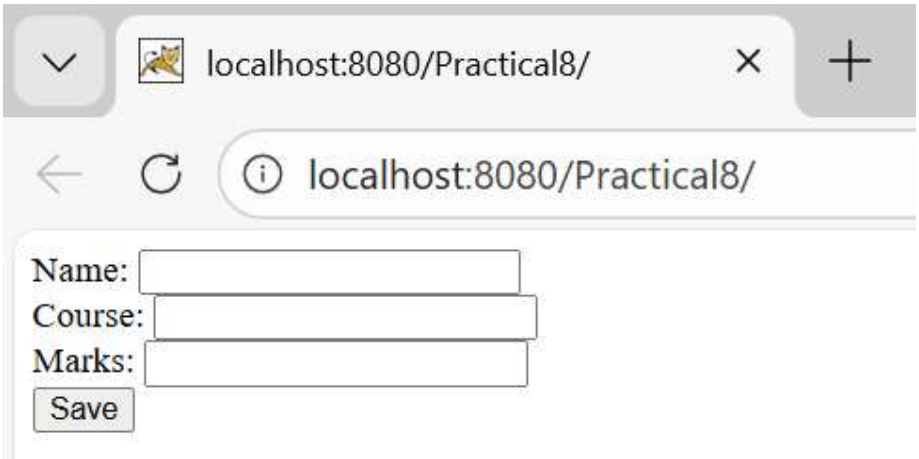
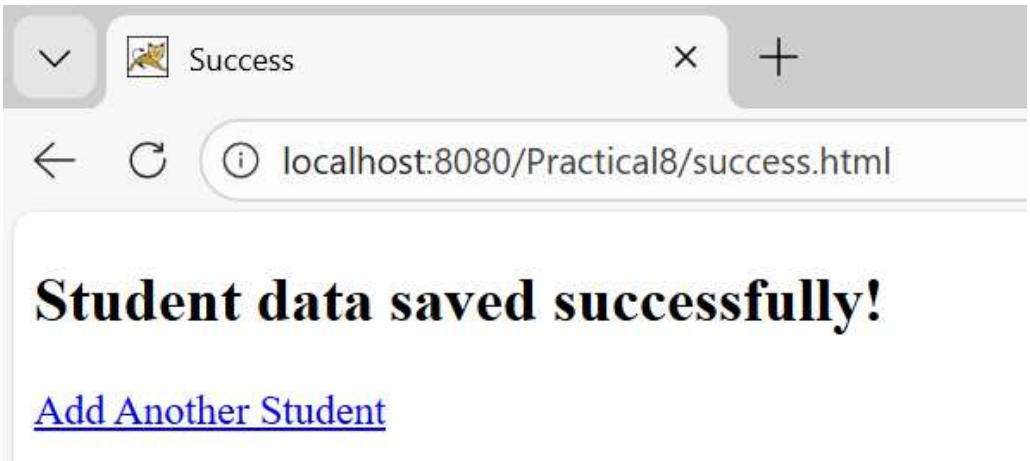
```
<servlet-name>StudentServlet</servlet-name>
<servlet-class>StudentServlet</servlet-class>
</servlet>
<servlet-mapping>
<servlet-name>StudentServlet</servlet-name>
<url-pattern>/StudentServlet</url-pattern>
</servlet-mapping>
</web-app>
```

OUTPUT:






```
mysql> select * from student;
+-----+--------------+--------+-------+
| sid | name         | course | marks |
+-----+--------------+--------+-------+
|   3 | Mamta Kurdia | BCA    |    90 |
+-----+--------------+--------+-------+
1 row in set (0.00 sec)
```

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SYBCA "A"

PRACTICAL NO 9

1. Design a JDBC application using JSP to demonstrate the concept of ResultSetMetaData.

CODE:

ResultSetMetaData.jsp

```
<%@ page import="java.sql.*" %>
<html>
<head>
   <title>ResultSetMetaData Demo</title>
</head>
<body>
<h2>Student Table using ResultSetMetaData</h2>
<%
Connection con = null;
try {
   Class.forName("com.mysql.cj.jdbc.Driver");
   con = DriverManager.getConnection(
      "jdbc:mysql://localhost:3306/jdbc_demo", "root", "mamta123"
   );
   Statement st = con.createStatement();
   ResultSet rs = st.executeQuery("SELECT * FROM student");
   ResultSetMetaData rsmd = rs.getMetaData();
   int columnCount = rsmd.getColumnCount();
   out.println("<table border='1'>");
   out.println("<tr>");
   for(int i=1; i<=columnCount; i++) {
      out.println("<th>" + rsmd.getColumnName(i) + "</th>");
   }
   out.println("</tr>");
   while(rs.next()) {
      out.println("<tr>");
      for(int i=1; i<=columnCount; i++) {
         out.println("<td>" + rs.getString(i) + "</td>");
      }
      out.println("</tr>");
   }
   out.println("</table>");
```

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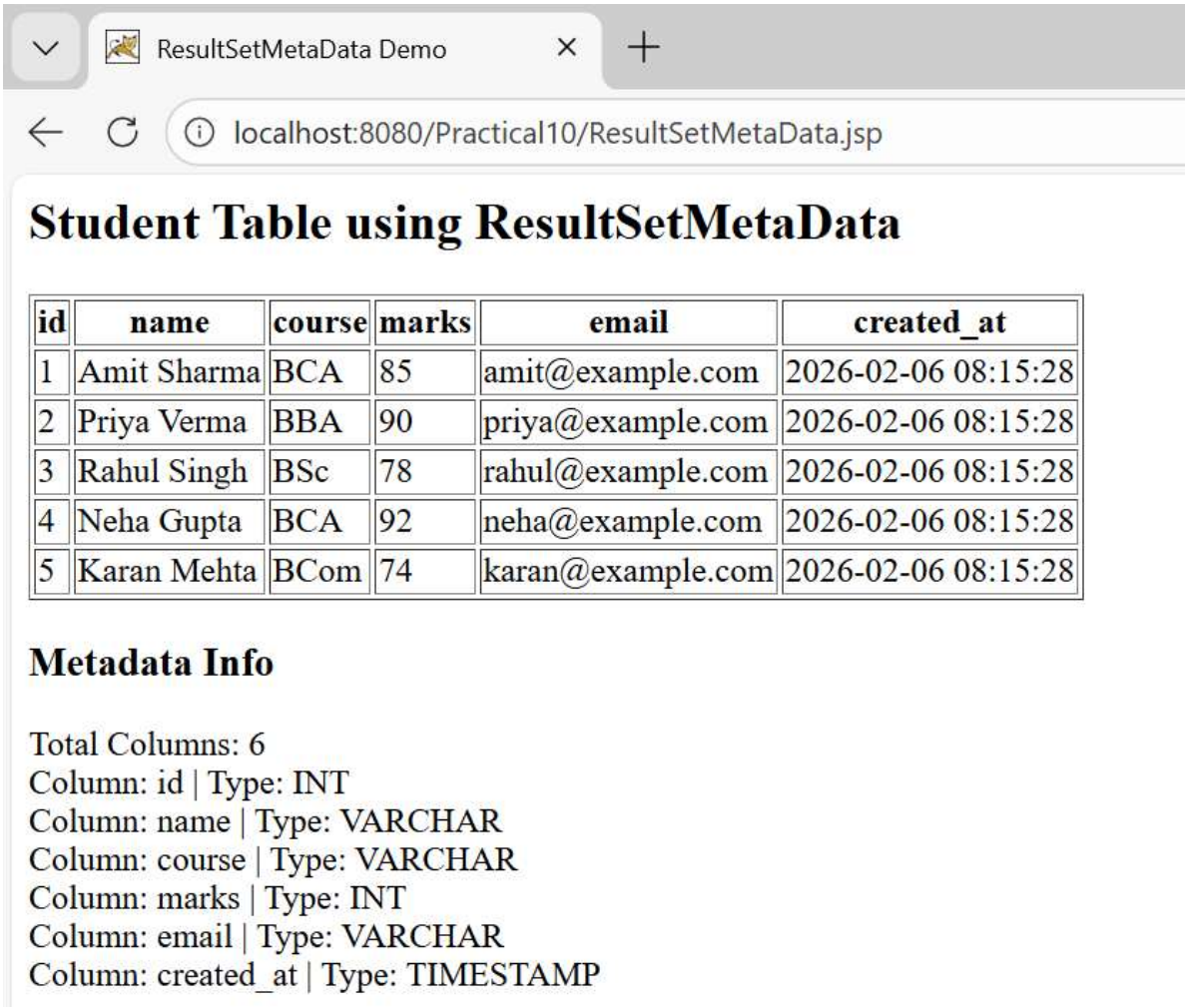
```
    out.println("<h3>Metadata Info</h3>");
    out.println("Total Columns: " + columnCount + "<br>");

    for(int i=1; i<=columnCount; i++) {
        out.println("Column: " + rsmd.getColumnName(i)
            + " | Type: " + rsmd.getColumnTypeName(i) + "<br>");
    }
} catch(Exception e) {
    out.println(e);
}
%>
</body>
</html>
```

OUTPUT:



Student Table using ResultSetMetaData

id	name	course	marks	email	created_at
1	Amit Sharma	BCA	85	amit@example.com	2026-02-06 08:15:28
2	Priya Verma	BBA	90	priya@example.com	2026-02-06 08:15:28
3	Rahul Singh	BSc	78	rahul@example.com	2026-02-06 08:15:28
4	Neha Gupta	BCA	92	neha@example.com	2026-02-06 08:15:28
5	Karan Mehta	BCom	74	karan@example.com	2026-02-06 08:15:28

Metadata Info

Total Columns: 6
Column: id | Type: INT
Column: name | Type: VARCHAR
Column: course | Type: VARCHAR
Column: marks | Type: INT
Column: email | Type: VARCHAR
Column: created_at | Type: TIMESTAMP

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Roll no: 5855
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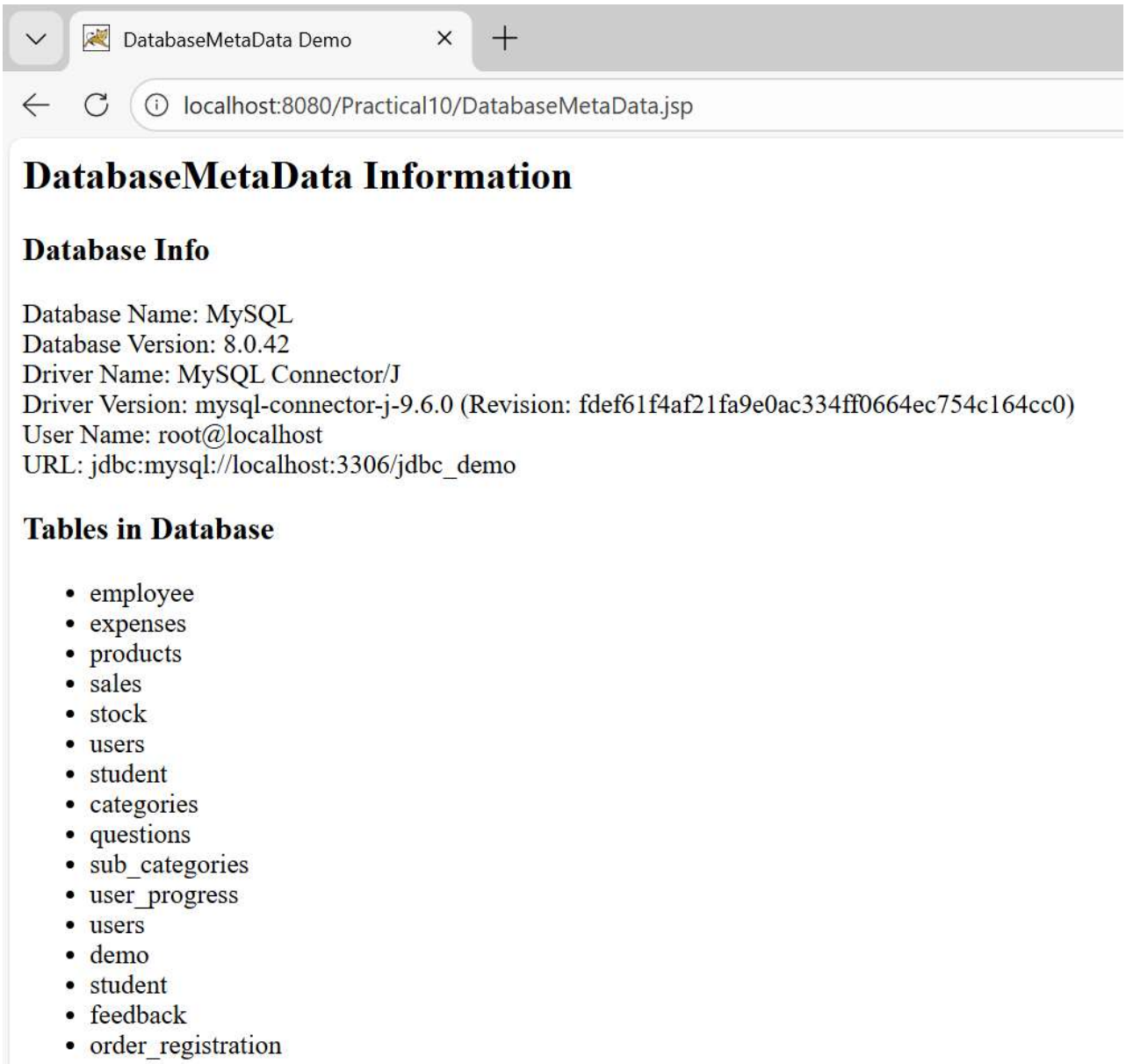
2. Design a JDBC application using JSP to demonstrate the concept of DatabaseMetaData.

CODE:

DatabaseMetaData.jsp

```
<%@ page import="java.sql.*" %>
<html>
<head>
   <title>DatabaseMetaData Demo</title>
</head>
<body>
<h2>DatabaseMetaData Information</h2>
<%
Connection con = null;
try {
   Class.forName("com.mysql.cj.jdbc.Driver");
   con = DriverManager.getConnection(
      "jdbc:mysql://localhost:3306/jdbc_demo", "root", "mamta123"
   );
   DatabaseMetaData dbmd = con.getMetaData();
   out.println("<h3>Database Info</h3>");
   out.println("Database Name: " + dbmd.getDatabaseProductName() + "<br>");
   out.println("Database Version: " + dbmd.getDatabaseProductVersion() + "<br>");
   out.println("Driver Name: " + dbmd.getDriverName() + "<br>");
   out.println("Driver Version: " + dbmd.getDriverVersion() + "<br>");
   out.println("User Name: " + dbmd.getUserName() + "<br>");
   out.println("URL: " + dbmd.getURL() + "<br>");
   out.println("<h3>Tables in Database</h3>");
   ResultSet tables = dbmd.getTables(null, null, "%", new String[]{"TABLE"});
   out.println("<ul>");
   while(tables.next()) {
      out.println("<li>" + tables.getString("TABLE_NAME") + "</li>");
   }
   out.println("</ul>");
} catch(Exception e) {
   out.println(e);
}
%>
```

```
</body>
</html>
```

OUTPUT: